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Distinguishing Carcinoma-Associated Fibroblasts from Normal Fibroblasts in Non-Small Cell Lung Cancer via Gene Expression and Functional Profiling

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1. Abstract

Carcinoma-associated fibroblasts (CAFs) are key drivers of non-small cell lung cancer (NSCLC) progression, yet distinguishing them from normal fibroblasts (NFs) remains difficult due to subtle transcriptional differences. In this study, publicly available gene expression data were analyzed using exploratory visualization, unsupervised clustering, and supervised classification approaches. While unsupervised methods failed to clearly separate CAFs from NFs, supervised models, particularly Random Forest and Linear Discriminant Analysis, effectively identified informative expression patterns. Functional enrichment analyses consistently revealed activation of extracellular matrix remodeling, developmental signaling, and oncogenic pathways in CAFs. These findings provide deeper insight into CAF biology and suggest potential therapeutic targets in NSCLC.

2. Introduction

Non-small cell lung cancer (NSCLC) is the most prevalent form of lung cancer and a leading cause of cancer-related mortality worldwide. Increasing evidence highlights the tumor microenvironment (TME) as a critical contributor to cancer progression and therapeutic resistance. Within the TME, carcinoma-associated fibroblasts (CAFs) have emerged as key players in promoting tumor growth, remodeling the extracellular matrix, and modulating immune responses. However, distinguishing CAFs from normal fibroblasts (NFs)

based on gene expression remains challenging due to overlapping transcriptional profiles and cellular heterogeneity. In this study, publicly available gene expression data from matched CAFs and NFs isolated from NSCLC tumors and adjacent healthy tissue were analyzed to investigate their molecular differences. A combination of computational and statistical methods was applied, including exploratory data visualization, unsupervised clustering, and supervised classification techniques. Additionally, functional enrichment and pathway analyses were performed to identify biological processes associated with CAF-specific phenotypes. This integrative approach aims to assess the ability of computational tools to distinguish CAFs from NFs and to identify biologically relevant pathways associated with CAFs in NSCLC.

3. Materials and Methods

3.1. Dataset

The dataset of matched carcinoma-associated fibroblasts (CAFs) and normal fibroblasts (NFs) was obtained from the Gene Expression Omnibus (GEO) under accession ID GSE22862 [1], with gene expression profiles generated using microarray technology. Data were downloaded using the getGEO function from the GEOquery R package [2], enabling extraction of both the expression matrix and sample metadata. Samples were classified as tumoral (T) or normal (N) based on metadata annotations, providing clear stratification into the two groups. Data quality and comparabil-

ity were assessed by inspecting boxplots of expression value distributions across samples. As the data were already log2-transformed and showed consistent distributions, no additional normalization was required.

3.2. Principal Component Analysis

Principal Component Analysis (PCA)[3] is an unsupervised dimensionality reduction technique used to identify patterns in high-dimensional datasets by transforming the data into a set of orthogonal components (principal components) that capture the greatest variance.

3.3. K-means Clustering

K-means clustering[4] is an unsupervised machine learning algorithm used to partition a dataset into a predefined number of clusters (k), based on feature similarity. The algorithm iteratively assigns each data point to the nearest cluster centroid and then recalculates the centroids based on the current cluster members. This process continues until convergence, typically when cluster assignments no longer change or the change falls below a defined threshold. K-means aims to minimize the within-cluster sum of squares, thereby ensuring that points within each cluster are as similar as possible to each other and as different as possible from those in other clusters.

3.4. Random Forest

Random Forest[5] is a supervised ensemble learning method used for classification and regression tasks. It operates by constructing a large number of decision trees during training and outputting the class (for classification) or mean prediction (for regression) based on the majority vote or average across individual trees. Each tree is built from a bootstrap sample of the data, and at each split, a random subset of features is considered to introduce diversity among the trees. This approach improves predictive accuracy and reduces the risk of overfitting compared to a single decision tree.

3.5. Linear Discriminant Analysis

Linear Discriminant Analysis (LDA)[6] is a supervised classification method that projects high-dimensional data onto a lower-dimensional space while maximizing class separability. It does this by finding linear combinations of features, called discriminant functions, that best separate the predefined classes. LDA assumes that the data from each class are normally distributed and share a common covariance matrix. It is particularly effective when class distributions are approxi-

mately Gaussian and the within-class variance is relatively small compared to between-class variance.

3.6. LASSO Logistic Regression

Lasso Logistic Regression[7] is a regularized form of logistic regression that incorporates an L1 penalty term to perform variable selection and reduce model complexity. By adding the absolute values of the regression coefficients to the loss function, the Lasso penalty encourages sparsity in the model shrinking some coefficients to exactly zero. This makes it particularly useful for high-dimensional datasets where many features may be irrelevant or correlated. The regularization strength is controlled by a tuning parameter, typically selected through cross-validation to balance model fit and generalizability.

3.7. g:Profiler

The g:Profiler R package[8] is a tool used to interpret gene lists by identifying significantly enriched biological terms and pathways. It enables users to connect gene sets to functional categories such as Gene Ontology (GO) terms, KEGG and Reactome pathways, transcription factor targets, and protein complexes. The package sends queries to the g:Profiler web service, performs enrichment analysis using a hypergeometric test, and adjusts for multiple testing using procedures like Benjamini–Hochberg. This allows for systematic biological interpretation of gene sets, such as those derived from differential expression or clustering analyses, within the R environment.

3.8. PathfindR

The PathfindR R package[9] is used for active sub-network enrichment analysis, integrating gene expression or other omics data with known biological interaction networks. It identifies functionally relevant pathways by detecting active subnetworks, groups of interconnected genes, within a protein-protein interaction network that are significantly enriched for differentially expressed genes. These subnetworks are then mapped to pathways from databases such as KEGG, Reactome, or BioCarta. PathfindR applies enrichment analysis iteratively and ranks pathways based on significance, providing a more context-aware interpretation of biological processes compared to traditional overrepresentation methods.

4. Results

4.1. Exploratory analysis

To gain an initial overview of the dataset, exploratory analyses were conducted using boxplots and principal component analysis (PCA). Boxplots of normalized gene expression values across all samples (Figure 1) showed consistent median expression levels and similar interquartile ranges, indicating good data quality and no apparent batch effects. While minor variability in outlier counts was observed, mainly at the upper expression range, the overall distribution appeared stable across both carcinoma-associated fibroblasts (CAFs, "T") and normal fibroblasts (NFs, "N"), supporting the suitability of the dataset for downstream analysis.

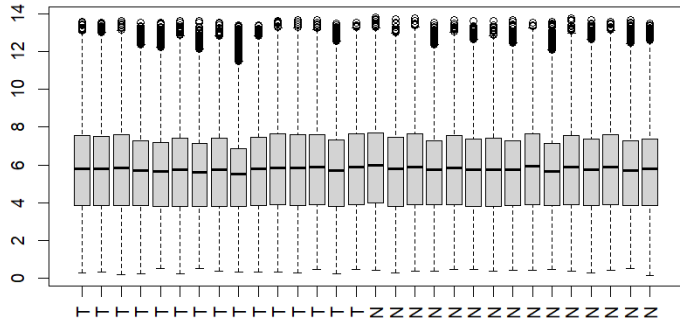


Figure 1: Boxplot of gene expression values across all samples.

PCA was performed to explore global expression patterns and assess natural group separation Figure 2. The visualization of the first two principal components

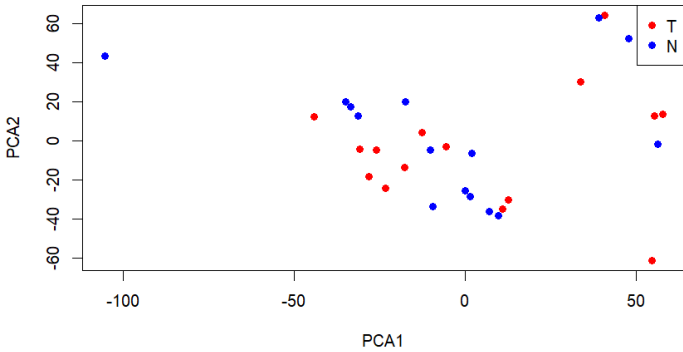


Figure 2: PCA of components 1 & 2. Red dots are the CAFs samples, while blue dots are the NFs samples

revealed substantial overlap between CAFs and NFs samples, with no clear clustering by group. This indicate that any distinguishing expression patterns between the two fibroblast types are likely subtle and require supervised methods for effective classification.

4.2. Unsupervised clustering

To further assess the potential for natural group separation, unsupervised K-means clustering was per-

formed with $k = 2$, corresponding to the two biological groups: CAFs ("T") and NFs ("N") (Figure 3). Clus-

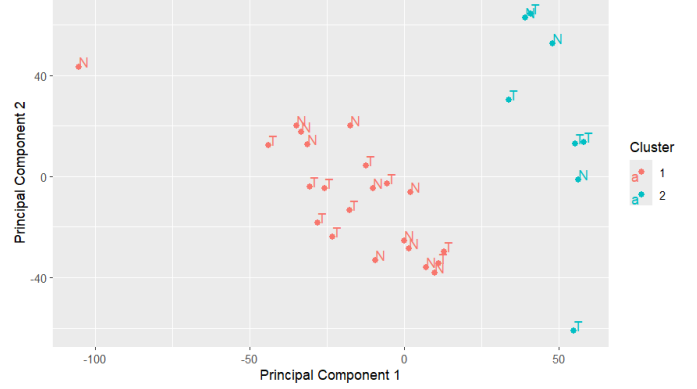


Figure 3: K-means clustering ($k = 2$) of gene expression profiles. Samples are colored by cluster assignment: Cluster 1 (coral) and Cluster 2 (turquoise).

ter assignments showed poor alignment with known sample labels, as both CAF and NF samples were distributed across clusters with substantial overlap. These findings, together with the PCA results, indicate that unsupervised approaches, which rely on dominant global patterns, are insufficient to distinguish CAFs from NFs.

4.3. Supervised Classification

To evaluate whether supervised learning methods could better capture the differences between CAFs and NFs, three classification models were applied: Random Forest (RF), LASSO logistic regression, and Linear Discriminant Analysis (LDA).

4.3.1 Random Forest

A Random Forest classifier was trained using expression data from all genes and 1,800 trees. The out-of-bag (OOB) error rate stabilized at 26.67%, indicating moderate classification performance between CAFs and NFs, Figure 4. Feature importance anal-

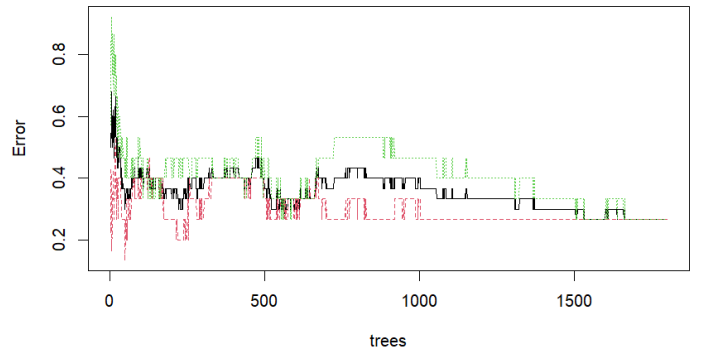


Figure 4: Out-of-bag (OOB) error rate across the number of trees in the Random Forest model.

ysis highlighted a subset of genes with strong discrim-

inatory power. A heatmap of the top 25 most important genes (Figure 5) revealed clear expression differences between groups. Despite some heterogeneity, the heatmap showed partial separation between CAF and NF samples, with several genes displaying consistent group-specific patterns (e.g., lower expression in N-labeled columns marked by blue blocks). These genes may serve as biologically relevant markers contributing to the classifier’s performance.

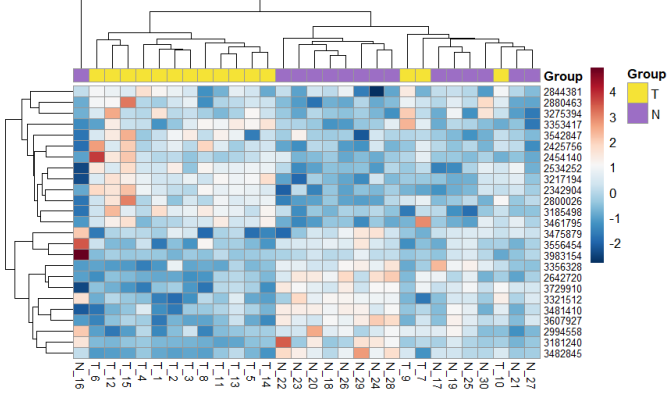


Figure 5: Heatmap of the top 25 most important genes identified by the Random Forest classifier. Rows correspond to genes and columns to samples, with expression values scaled per gene (row-wise z-scores). Red indicates higher-than-average expression and blue indicates lower-than-average expression. Hierarchical clustering of both genes and samples reveals partial separation between CAFs ("T") and NFs ("N").

4.3.2 Linear discriminant analysis

Linear Discriminant Analysis (LDA) was applied to a subset of genes preselected using a t-test ($p < 0.1$) to reduce noise and focus on informative features. The dataset was split into class-balanced training and test sets to ensure equal representation of CAFs and NFs. On the training set, LDA achieved clear group separation along the first discriminant axis, indicating effective feature extraction. When tested on the independent test set, the model correctly classified 9 out of 10 samples. The corresponding ROC curve, Figure 6, showed strong discriminatory performance, with an area under the curve (AUC) of 0.9. Despite expression heterogeneity, LDA successfully captured key variance patterns that distinguish CAFs from NFs.

4.3.3 LASSO logistic regression

To evaluate the predictive power of gene expression data, LASSO logistic regression was used to distinguish CAFs from NFs. The dataset was split into class-balanced training and test sets to ensure equal representation of both groups. The model was trained on the training set, with 10-fold cross-validation used to select the optimal regularization pa-

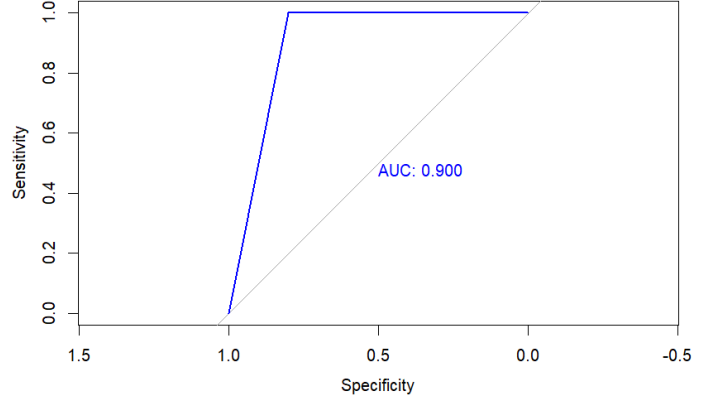


Figure 6: ROC curve for LDA classification of CAFs versus NFs on the test set. The x-axis represents specificity, and the y-axis represents sensitivity. The area under the curve (AUC) of 0.9 reflects strong classification performance.

rameter (lambda). On the independent test set, the model correctly classified 8 out of 10 samples. The ROC curve (Figure 7) demonstrated strong performance, with an area under the curve (AUC) of 0.88, indicating high sensitivity and specificity in distinguishing CAFs from NFs.

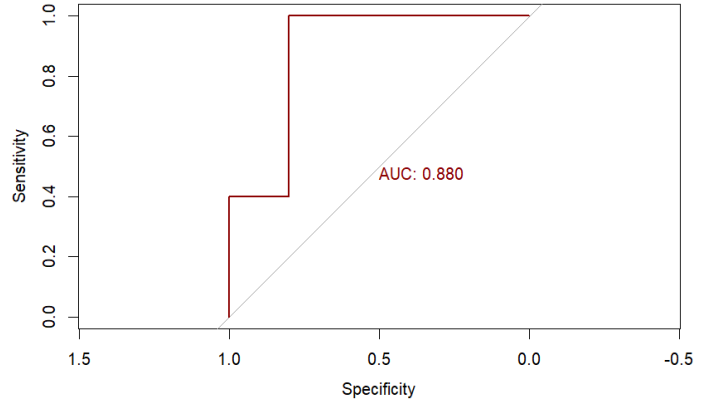


Figure 7: ROC curve for LASSO logistic regression on the test set. The x-axis shows specificity and the y-axis shows sensitivity. The area under the curve (AUC) of 0.88 reflects strong classification performance.

4.3.4 Model Comparison

To compare the predictive performance of the three classifiers, 5-fold cross-validation was applied to Random Forest (RF), Linear Discriminant Analysis (LDA), and LASSO logistic regression using the filtered gene expression dataset. As shown in Figure 8, Random Forest achieved the highest average accuracy, 0.93, with moderate variability, indicating strong and consistent performance. LDA followed with a mean accuracy of 0.87 and slightly lower variance, suggesting reliable class separation despite the model’s linear assumptions. LASSO logistic regression yielded the lowest average accuracy, 0.70, and the highest variability, reflecting its sensitivity to sample composition

and regularization parameters. To obtain more robust

ple composition and regularization settings.

4.4. Enrichment analysis

4.4.1 g:Profiler

To gain functional insight into the gene expression differences between CAFs and NFs, Gene Ontology (GO) enrichment analysis was performed using the gprofiler2 package. The analysis integrated multiple annotation sources to identify significantly overrepresented biological processes and pathways. A visual summary of enriched terms across databases is shown in Figure 10.

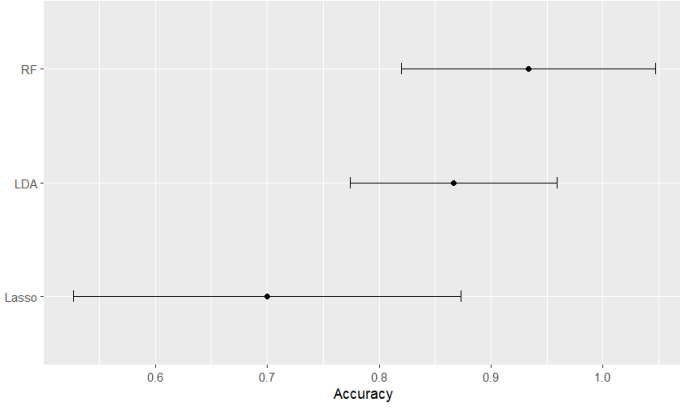


Figure 8: Model comparison based on 5-fold cross-validation accuracy. The dot represents the mean classification accuracy for each model (Random Forest, LDA, LASSO), and the horizontal lines indicate the range (minimum and maximum) across folds.

and generalizable performance estimates, the classifiers were further evaluated using repeated 5×5 -fold cross-validation, which reduces variability due to random data partitioning. As shown in Figure 9, Random Forest again outperformed the other models, achieving a mean accuracy of 0.88 with moderate variability, reinforcing its robustness to expression heterogeneity. LASSO demonstrated improved performance, mean accuracy of 0.73, and reduced variance relative to its single 5-fold CV results, suggesting increased stability under repeated sampling. LDA showed a slight decrease in accuracy from 0.87 to 0.85 but maintained consistent performance overall. Overall, the results

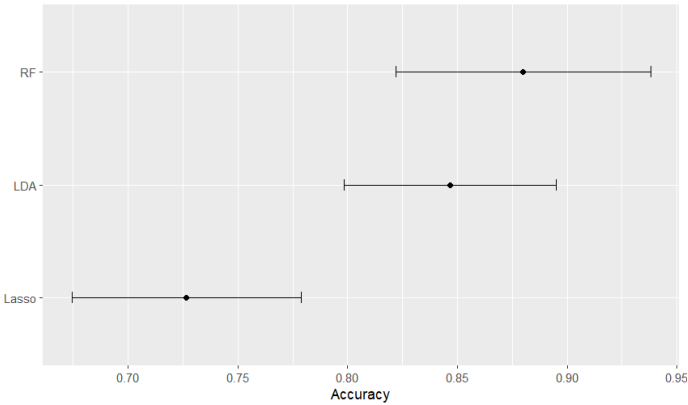


Figure 9: Model comparison based on repeated 5×5 -fold cross-validation accuracy. Dots indicate the mean classification accuracy over repeated folds for each model, while horizontal lines show the variability across runs.

highlight the strong and relatively stable performance of both Random Forest and LDA in classifying CAFs and NFs based on gene expression data. Although Random Forest achieved the highest accuracy, both models exhibited only minor sensitivity to repeated data partitioning. In contrast, LASSO showed greater variability, suggesting it may be more affected by sam-

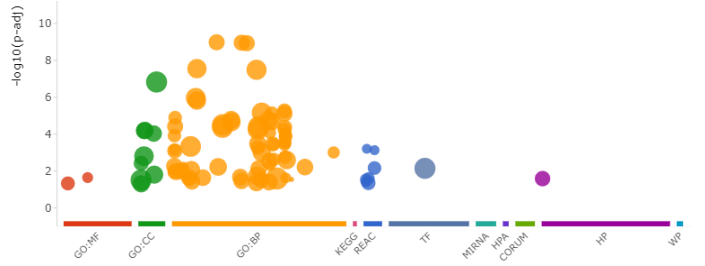


Figure 10: Functional enrichment plot summarizing overrepresented terms among the top discriminative genes between CAFs and NFs. Terms are grouped by database (x-axis), including GO Biological Process (BP), Cellular Component (CC), Molecular Function (MF), Reactome (REAC), Human Phenotype Ontology (HP), transcription factor motifs (TF), KEGG pathways, miRNA targets, Human Protein Atlas (HPA), CORUM protein complexes, and WikiPathways (WP). Each dot represents an enriched term, and the y-axis displays the statistical significance as $-\log_{10}(p\text{-value})$.

The results revealed strong and statistically significant enrichment of terms related to extracellular matrix (ECM) organization and tissue development (see Appendix, Table 1), underscoring the central role of ECM remodeling and developmental signaling in CAF biology. Notably, GO terms associated with renal and epithelial morphogenesis, such as *nephron development* and *kidney morphogenesis*, were also significantly enriched, suggesting a reactivation of embryonic programs within the tumor-associated stroma. Enriched cellular component terms such as *collagen-containing extracellular matrix* and *cell junctions*, along with molecular function terms like *extracellular matrix structural constituent* and *metalloendopeptidase activity*, pointed to active extracellular matrix turnover. While these terms were less statistically significant than the top biological processes, they consistently point to the structural and enzymatic remodeling activities characteristic of CAFs. Together, these results offer a biologically coherent interpretation of the transcriptomic differences between CAFs and NFs and support a model in which CAFs promote tumor progression through stromal reprogramming and dynamic ECM regulation.

4.4.2 PathfindR

To further characterize the biological programs distinguishing CAFs from NFs, pathway enrichment analysis was conducted using the pathfindR package across multiple databases.

Reactome-based analysis revealed strong enrichment for pathways involved in collagen biosynthesis, extracellular matrix (ECM) organization, and post-translational modifications. Notable terms, including *Collagen biosynthesis and modifying enzymes*, *Collagen formation*, and *Extracellular matrix organization*, highlight ECM remodeling as a defining characteristic of CAF biology. Enrichment of glycosylation-related pathways, such as *O-glycosylation of TSR domain-containing proteins* and *Diseases associated with O-glycosylation of proteins*, suggests active regulation of ECM maturation and structural stability. Additionally, the involvement of developmental and oncogenic signaling pathways, including *NOTCH* and *RET signaling*, points to broader activation of tumor-promoting transcriptional programs. All pathways showed strong statistical significance (Figure 11). A

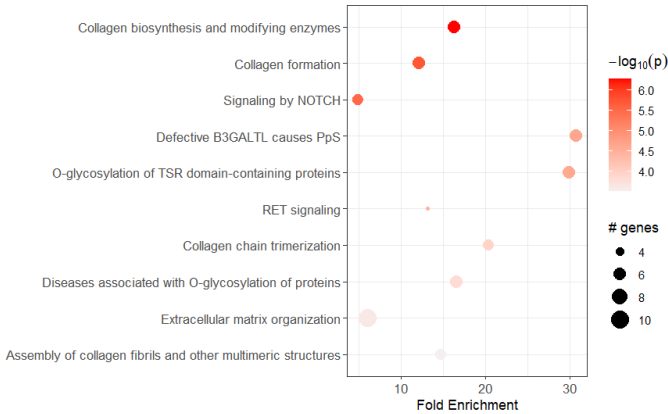


Figure 11: Reactome pathway enrichment analysis of genes distinguishing carcinoma-associated fibroblasts (CAFs) from normal fibroblasts (NFs). The dot plot displays the top significantly enriched pathways, with pathway terms on the y-axis and fold enrichment on the x-axis. Statistical significance is represented by dot color ($-\log_{10}(p)$), and dot size corresponds to the number of genes associated with each term.

term-gene network visualization (Figure 12) showed overlapping gene contributors across ECM-related pathways, particularly genes from the COL (collagen) and ADAMTS protease families. This underscores a coordinated enrichment of structural, regulatory, and signaling components involved in matrix remodeling. These findings reinforce the central role of CAFs in stromal reorganization and tumor progression. Full pathway details are provided in Appendix Table 2.

Gene Ontology Biological Process (GO-BP) analysis identified *positive* and *negative regulation of miRNA transcription* as the most significantly enriched terms,

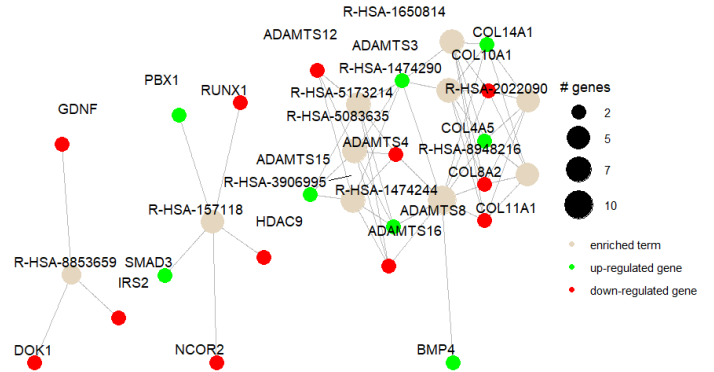


Figure 12: Term-gene network visualization. Beige nodes represent enriched pathway terms (top 10 terms), with node size proportional to the number of genes involved in each pathway. Green nodes indicate upregulated genes, and red nodes indicate downregulated genes. Edges link genes to the pathways in which they are involved, illustrating shared gene contributions across multiple biological processes.

pointing to a central role for transcriptional control via miRNA networks in CAF biology. Additional moderately significant processes, including *axonogenesis*, *TGF-beta receptor signaling*, *cellular response to BMP stimulus*, *glucose import*, *glomerular filtration*, *cytokine regulation*, and *DNA repair*, highlight the broad transcriptional reprogramming and functional plasticity that characterize CAFs. These findings are visualized in Figure 13.

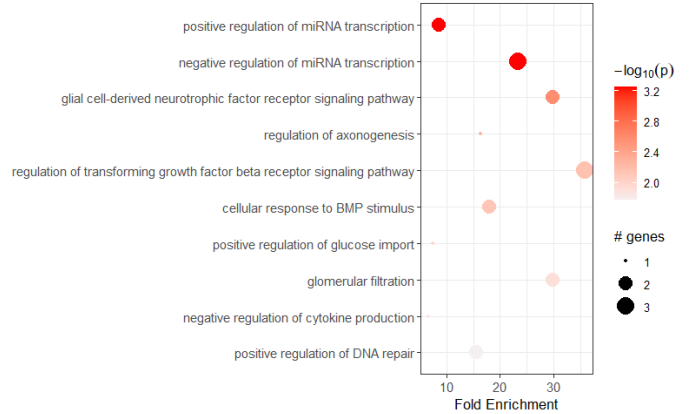


Figure 13: GO-BP pathway enrichment analysis of genes distinguishing CAFs from NFs. The dot plot displays the top significantly enriched pathways, with pathway terms on the y-axis and fold enrichment on the x-axis. Statistical significance is represented by dot color ($-\log_{10}(p)$), and dot size corresponds to the number of genes associated with each term.

Finally, Gene Ontology Cellular Component (GO-CC) analysis revealed statistically significant enrichment for *endoplasmic reticulum lumen* and *adherens junction*, indicating active protein processing and cell-cell adhesion in CAFs. Additional moderately significant terms, including *transcription regulator complex*, *chromatin*, *apical plasma membrane* and *spindle pole*, suggest extensive changes in transcriptional regulation, secretion, membrane organization, and cy-

toskeletal architecture. As shown in Figure 14.

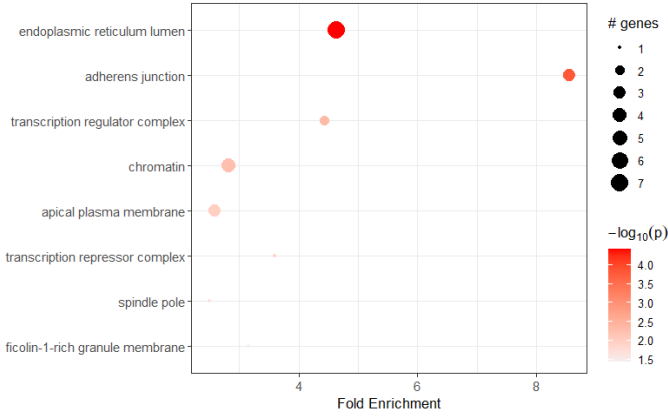


Figure 14: GO-CC pathway enrichment analysis of genes distinguishing CAFs from NFs. The dot plot displays the top significantly enriched pathways, with pathway terms on the y-axis and fold enrichment on the x-axis. Statistical significance is represented by dot color ($-\log_{10}(p\text{-value})$), and dot size corresponds to the number of genes associated with each term.

Together, these pathway enrichment analyses reveal a complex remodeling phenotype in CAFs, marked by coordinated activation of ECM biosynthesis, transcriptional reprogramming—particularly via miRNA regulation and structural adaptations involving secretion, adhesion, and cytoskeletal organization, all of which underscore the multifaceted role of CAFs in tumor progression.

5. Discussion

This study explored gene expression differences between carcinoma-associated fibroblasts (CAFs) and normal fibroblasts (NFs). Unsupervised methods such as PCA and K-means clustering were unable to clearly distinguish the two cell types. However, supervised approaches—particularly Random Forest and Linear Discriminant Analysis (LDA) successfully identified subtle but informative transcriptional patterns that differentiate CAFs from NFs. To interpret the biological relevance of these differences, two complementary enrichment tools were used: g:Profiler and pathfindR. Despite differences in methodology, both consistently highlighted biological themes characteristic of CAFs, including extracellular matrix (ECM) organization, collagen biosynthesis, and developmental signaling. These findings align well with known features of CAF biology, but each tool offered unique strengths. g:Profiler used classical overrepresentation analysis across multiple annotation databases, offering a broad overview of enriched biological processes. Key findings included upregulation of ECM organization, kidney/nephron development, and morphogenesis, suggesting that CAFs may reactivate embryonic programs to remodel the tumor microenvironment.

Enrichment of cellular components like collagen-containing ECM and cell junctions further supported a stroma-driven remodeling phenotype. However, g:Profiler’s approach lacked context on how enriched terms interact functionally. In contrast, pathfindR employed an active subnetwork-based approach, integrating protein-protein interaction networks to identify functionally coherent modules. This method provided more mechanistic insight, highlighting pathways such as collagen formation, ECM modification, and post-translational O-glycosylation. Importantly, it also uncovered oncogenic pathways like NOTCH and RET signaling, and transcriptional regulators involved in miRNA biogenesis. These features were not emphasized by g:Profiler. Visualizations from pathfindR further illustrated shared gene contributors, COL and ADAMTS families, across pathways, reinforcing the central role of matrix regulation and signaling reprogramming in CAFs. Across both tools, a consistent narrative emerged: CAFs exhibit a transcriptional signature enriched in extracellular matrix remodeling, developmental reprogramming, and signaling plasticity. The altered regulation of genes involved in collagen synthesis and metalloprotease activity indicates active ECM deposition and remodeling, which are key features of tumor-promoting stroma. Concurrent enrichment of morphogenesis and renal development terms suggests reactivation of embryonic pathways that facilitate tissue invasion and remodeling. Furthermore, the involvement of miRNA-related transcriptional regulation and oncogenic signaling pathways (e.g., NOTCH, RET) underscores the adaptive and dynamic nature of CAFs in response to tumor-derived cues. Combining g:Profiler’s broad functional landscape with pathfindR’s mechanistic depth offered a comprehensive view of the molecular distinctions between CAFs and NFs. This integrative approach underscores the multifaceted role of CAFs in reshaping the tumor microenvironment.

5.1. Comparison with Previous Studies and Potential Applications

Previous research has linked CAF gene expression profiles to poor prognosis in non-small cell lung cancer (NSCLC). For instance, an earlier study identified an 11-gene prognostic signature validated across several datasets, with enrichment in focal adhesion and MAPK signaling, key pathways in tumor-stroma interactions. Furthermore, 14 out of 46 stroma-associated genes were differentially expressed in tumor versus normal tissue, and several were responsive to TGF- β 1, highlighting the role of tumor-derived factors in CAF reprogramming.

Although this study did not replicate that spe-

cific gene signature, it confirmed and extended key findings. Independent enrichment was observed in pathways related to extracellular matrix remodeling, developmental signaling (including BMP, TGF-*beta*, and nephrogenesis-associated pathways), and oncogenic cascades (such as NOTCH and RET), all of which are central to CAF function. Through the integration of machine learning-based gene prioritization and functional enrichment analysis, the study identified subtle yet meaningful transcriptional shifts, reinforcing the concept of CAFs as reprogrammed and reactive components of the tumor microenvironment. These insights have clear translational potential. ECM-related genes and signaling hubs identified here could serve as biomarkers for CAF presence or activation in patient samples. Additionally, they represent potential therapeutic targets, particularly for interventions aiming to disrupt stromal remodeling or CAF-tumor signaling feedback. The findings may also help refine prognostic models by incorporating CAF-derived features, improving risk stratification and treatment planning in NSCLC.

6. Conclusion

This study demonstrates the value of computational approaches in distinguishing carcinoma-associated fibroblasts (CAFs) from normal fibroblasts (NFs) in non-small cell lung cancer (NSCLC). Although unsupervised analyses failed to clearly distinguish between CAFs and NFs, supervised models, particularly Random Forest and Linear Discriminant Analysis, were able to accurately classify them revealing meaningful differences in gene expression patterns. Functional enrichment analyses consistently highlighted pathways related to extracellular matrix organization, collagen biosynthesis, and developmental signaling, supporting a role for CAFs in tumor progression through stromal remodeling and transcriptional reprogramming. In the end, this work underscores the power of machine learning and network-based tools to detect coordinated changes in the tumor microenvironment. While further validation is needed, these findings offer both a computational framework and biological insights that may inform future therapeutic strategies targeting CAFs in NSCLC.

A. Appendix

A.1. Enrichment Table from g:Profiler

Category	Term Name	ID	p-value	#Genes
GO:BP	Extracellular matrix organization	GO:0030198	1.05×10^{-9}	20
GO:BP	Extracellular structure organization	GO:0043062	1.11×10^{-9}	20
GO:BP	Tissue development	GO:0009888	2.82×10^{-8}	45
GO:BP	Renal system development	GO:0072001	7.79×10^{-6}	16
GO:BP	Kidney morphogenesis	GO:0060993	7.21×10^{-6}	10
GO:BP	Regulation of multicellular organismal process	GO:0051239	6.84×10^{-6}	51
GO:BP	Nephron development	GO:0072006	5.43×10^{-6}	12
GO:BP	Animal organ morphogenesis	GO:0009887	1.53×10^{-6}	29
GO:CC	Cell periphery	GO:0071944	1.49×10^{-7}	85
GO:CC	Extracellular matrix	GO:0031012	6.30×10^{-5}	18
GO:CC	External encapsulating structure	GO:0030312	6.47×10^{-5}	18
GO:CC	Collagen-containing extracellular matrix	GO:0062023	9.32×10^{-5}	15
GO:CC	Adherens junction	GO:0005912	3.83×10^{-3}	9
GO:CC	Cell junction	GO:0030054	1.56×10^{-3}	38
Reactome	Defective B3GALTL causes Peters-plus syndrome	R-HSA-5083635	6.22×10^{-4}	6
Reactome	O-glycosylation of TSR domain-containing proteins	R-HSA-5173214	7.41×10^{-4}	6
Reactome	O-linked glycosylation	R-HSA-5173105	6.89×10^{-3}	8
Reactome	Collagen biosynthesis	R-HSA-1650814	2.93×10^{-2}	6
Reactome	ECM degradation	R-HSA-1474228	4.45×10^{-2}	8
TF Motif	MOVO-B (GNGGGGG)	TF:M01104.1	7.17×10^{-3}	78
HP	Narrow mouth	HP:0000160	2.59×10^{-2}	11
GO:MF	ECM structural constituent (tensile)	GO:0030020	2.27×10^{-2}	5
GO:MF	Metalloendopeptidase activity	GO:0004222	4.70×10^{-2}	7

Table 1: Summary of selected enriched terms from g:Profiler functional enrichment analysis performed on the top differentially expressed genes between carcinoma-associated fibroblasts (CAFs) and normal fibroblasts (NFs). Each row lists the functional category (e.g., GO Biological Process, Cellular Component, Molecular Function, Reactome, etc.), the name and identifier of the enriched term, the statistical significance of enrichment (adjusted p-value), and the number of genes from the input list associated with that term.

Category	Term Name	ID	p-value	Up-regulated Genes	Down-regulated Genes
Reactome	Collagen biosynthesis and modifying enzymes	R-HSA-1650814	5.65e-04	SMAD3	KLF5
Reactome	Collagen formation	R-HSA-1474290	5.66e-04	BMP4, SMAD3	NCOR2
Reactome	Signaling by NOTCH	R-HSA-157118	2.64e-03		GDNF, SULF1
Reactome	Defective B3GALTL causes PpS	R-HSA-5083635	5.49e-03		SIPA1L1
Reactome	O-glycosylation of TSR domain-containing proteins	R-HSA-5173214	6.99e-03	BCL9, SMAD3	LATS2
Reactome	RET signaling	R-HSA-8853659	7.59e-03	BMP4	ADAMTS12
Reactome	Collagen chain trimerization	R-HSA-8948216	1.10e-02		IRS2
Reactome	Diseases associated with O-glycosylation of proteins	R-HSA-3906995	1.23e-02		MYO1E, SULF1
Reactome	Extracellular matrix organization	R-HSA-1474244	1.40e-02		HDAC9
Reactome	Assembly of collagen fibrils and other multimeric structures	R-HSA-2022090	1.68e-02	FGF10	NPAS2
GO:BP	Positive regulation of miRNA transcription	GO:1902895	5.65e-04	SMAD3	KLF5
GO:BP	Negative regulation of miRNA transcription	GO:1902894	5.66e-04	SMAD3, NCOR2	BMP4
GO:BP	Glial cell-derived neurotrophic factor receptor signaling pathway	GO:0035860	2.64e-03	GDNF, SULF1	
GO:BP	Regulation of axonogenesis	GO:0050770	5.49e-03	SIPA1L1	
GO:BP	Regulation of transforming growth factor beta receptor signaling pathway	GO:0017015	7.00e-03	BCL9, SMAD3	LATS2
GO:BP	Cellular response to BMP stimulus	GO:0071773	7.59e-03	BMP4	ADAMTS12
GO:BP	Positive regulation of glucose import	GO:0046326	1.10e-02	IRS2	
GO:BP	Glomerular filtration	GO:0003094	1.23e-02	MYO1E, SULF1	
GO:BP	Negative regulation of cytokine production	GO:0001818	1.40e-02	HDAC9	
GO:BP	Positive regulation of DNA repair	GO:0045739	1.68e-02	FGF10	NPAS2
GO:CC	Endoplasmic reticulum lumen	GO:0005788	4.00e-05	BMP4, COL4A5, COL14A1	COL8A2, COL10A1, COL11A1, GAS6
GO:CC	Adherens junction	GO:0005912	1.64e-04	MYO1E, PTPRM, MAGI1	–
GO:CC	Transcription regulator complex	GO:0005667	4.94e-03	SMAD3	HDAC9
GO:CC	Chromatin	GO:0000785	5.79e-03	SMAD3, ZNF711, BANF1	NCOR2
GO:CC	Apical plasma membrane	GO:0016324	1.05e-02	LMO7, SLC17A1, TCIRG1	–
GO:CC	Transcription repressor complex	GO:0017053	1.15e-02	NCOR2	–
GO:CC	Spindle pole	GO:0000922	2.39e-02	LATS2	–
GO:CC	Ficolin-1-rich granule membrane	GO:0101003	3.73e-02	TCIRG1	–

Table 2: Enriched pathways and terms identified by pathfindR across Reactome, GO Biological Process (BP), and GO Molecular Function (MF). Genes are split by their regulation status when available.

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